

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Voce Questions

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Viva Voce Questions	Assessment:	Assessment Tool 3– Viva Voce Questions
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B C Yugesh	Reg. No.:	192424282
Section / Batch:	2024	Date:	24/08/2026

Code of Conduct

I B C Yugesh (192424282) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B C Yugesh

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Introduction to Statistical Inference

1. What is statistical inference?

A. Statistical inference is the process of using sample data to draw conclusions or make predictions about a larger population.

2. Why is statistical inference important in data science?

A. It helps data scientists make decisions and predictions about a population without collecting data from every individual. It is useful for decision-making, prediction, and understanding patterns in data.

3. What is the difference between descriptive statistics and inferential statistics?

A. Descriptive statistics summarize and describe the data collected, using measures such as mean, median, and standard deviation. Inferential statistics use sample data to make conclusions or predictions about a larger population.

4. What is a population in statistics?

A. A population is the complete group of individuals, objects, or observations that a statistical study is interested in.

5. What is a sample?

A. A sample is a smaller subset selected from a population for the purpose of analysis.

6. Why do data scientists usually work with samples instead of complete populations?

A. Studying an entire population can be expensive, time-consuming, or

impossible. A properly selected sample can provide useful information about the population at a much lower cost.

7. What is a population parameter?

A. A population parameter is a numerical value that describes a characteristic of an entire population, such as the population mean or population proportion.

8. What is a sample statistic?

A. A sample statistic is a numerical value calculated from sample data, such as the sample mean, sample variance, or sample proportion.

9. Give a real-world example where statistical inference is used in industry.

A. A company may survey 1,000 customers to estimate the satisfaction level of all its customers. The sample results are used to make conclusions about the entire customer population.

10. What are the two major activities involved in statistical inference?

A. The two major activities are:

1. Estimation – estimating unknown population parameters.
2. Hypothesis testing – testing claims or assumptions about population parameters.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

A. The frequentist approach is a statistical method in which unknown population parameters are treated as fixed values, while the observed data are considered random.

12. What is the basic idea behind frequentist statistics?

A. The basic idea is to use repeated sampling and the long-run behavior of statistical procedures to make conclusions about unknown population parameters.

13. How does the frequentist approach differ from the Bayesian approach?

A. In the frequentist approach, parameters are fixed and probabilities describe the behavior of data over repeated samples. In the Bayesian approach, parameters are represented using probability distributions and prior knowledge can be combined with observed data.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

A. The population parameter is considered fixed, while the sample data are considered random because they can vary from one sample to another.

15. What does long-run frequency mean in frequentist statistics?

A. Long-run frequency refers to the proportion of times an event occurs when an experiment is repeated many times under the same conditions.

16. Why is random sampling important in the frequentist approach?

A. Random sampling helps ensure that the sample represents the population and allows probability theory to be used to understand sampling variability.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

A. No. In the frequentist approach, a population parameter is considered a fixed but unknown value. Randomness comes from the sample data rather than the parameter.

18. Give an example of a frequentist problem in a data science application.

A. A data scientist may want to determine whether a new recommendation algorithm increases the average number of purchases compared with the existing algorithm. Sample data can be used to test this claim statistically.

C. Point Estimation

19. What is a point estimate?

A. A point estimate is a single numerical value calculated from sample data and used to estimate an unknown population parameter.

20. What is an estimator?

A. An estimator is a statistical rule or formula used to calculate an estimate of a population parameter from sample data.

21. What is the difference between an estimator and an estimate?

A. An estimator is the formula or method used to estimate a parameter, while an estimate is the numerical result obtained after applying that estimator to a particular sample.

22. What is the point estimator for the population mean?

A. The sample mean is the point estimator for the population mean.

$$\bar{x} = \frac{\sum x_i}{n}$$

where n is the sample size.

23. How do you estimate a population proportion from a sample?

A. The population proportion is estimated using the sample proportion:

$$\hat{p} = \frac{x}{n}$$

where x is the number of successes and n is the sample size.

24. What is the point estimate of population variance?

A. The sample variance is commonly used as the point estimate of population variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

25. Why is the sample mean commonly used to estimate the population mean?

A. The sample mean is commonly used because it is unbiased for the population mean and generally becomes more accurate as the sample size increases.

26. What properties should a good estimator have?

A. A good estimator should ideally have:

- Unbiasedness
- Consistency
- Efficiency
- Low variance
- Reliability

27. What is an unbiased estimator?

A. An unbiased estimator is one whose expected value is equal to the true population parameter.

$$E(\hat{\theta}) = \theta$$

28. What is meant by the bias of an estimator?

A. Bias is the difference between the expected value of an estimator and the true population parameter.

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

If the bias is zero, the estimator is unbiased.

29. What is the difference between bias and variance of an estimator?

A. Bias measures how far the average estimate is from the true parameter.

Variance measures how much the estimates vary from sample to sample.

30. Why can two different samples produce different point estimates for the same population parameter?

A. Different samples contain different observations. Therefore, statistics such as the sample mean or sample proportion can vary from one sample to another due to sampling variability.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

A. A point estimate is calculated from a sample rather than the entire population. Since different samples can produce different estimates, the estimate has uncertainty.

32. What is the sampling distribution of an estimator?

A. The sampling distribution is the probability distribution of an estimator obtained from all possible samples of a given size from the population.

33. What is the standard error?

A. Standard error measures the variability of a statistic or estimator across repeated samples.

For a sample mean:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

34. How is standard error different from standard deviation?

A. Standard deviation measures the variability of individual observations within a dataset. Standard error measures the variability of a sample statistic, such as the sample mean, across repeated samples.

35. How does increasing the sample size affect the standard error of the sample mean?

A. Increasing the sample size decreases the standard error.

$$SE = \frac{s}{\sqrt{n}}$$

Therefore, a larger sample generally produces a more precise estimate.

36. Why does variability in an estimate decrease when the sample size increases?

A. With a larger sample, random differences between individual observations tend to balance out. Therefore, the sample estimate becomes more stable and closer to the population parameter.

37. What is the relationship between standard error and confidence intervals?

A. Standard error determines how much uncertainty is associated with an estimate. A larger standard error produces a wider confidence interval, while a smaller standard error produces a narrower confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

A. No. Different sample means are normal because different samples contain different observations. This difference is called sampling variability and does not necessarily mean that either sample is incorrect.

E. Confidence Intervals

39. What is a confidence interval?

A. A confidence interval is a range of values calculated from sample data that is used to estimate an unknown population parameter with a specified level of confidence.

40. Why is a confidence interval more informative than a point estimate?

A. A point estimate provides only one value, while a confidence interval provides a range and shows the uncertainty or precision associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

A. A 95% confidence level means that if the same sampling procedure were repeated many times, approximately 95% of the resulting confidence intervals would contain the true population parameter.

42. What is the difference between a confidence level and a confidence interval?

A. The confidence level is the percentage used to describe the reliability of the estimation procedure, such as 95%. The confidence interval is the actual numerical range calculated from the sample.

43. What factors determine the width of a confidence interval?

A. The width depends mainly on:

- Sample size
- Standard error or variability
- Confidence level
- Appropriate critical value from the statistical distribution

44. How does increasing the sample size affect the width of a confidence interval?

A. Increasing the sample size generally decreases the standard error, resulting in a narrower confidence interval and a more precise estimate.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

A. Increasing the confidence level from 95% to 99% makes the confidence interval wider because a higher level of confidence requires a larger range of possible values.

46. What is the difference between a z-confidence interval and a t-confidence interval?

A. A z-confidence interval uses the standard normal distribution and is

commonly used when the population standard deviation is known or the sample is sufficiently large. A t-confidence interval uses the t-distribution and is commonly used when the population standard deviation is unknown, especially with smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

A. The t-distribution is generally used when the population standard deviation is unknown and is estimated using the sample standard deviation, particularly when the sample size is small.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

A. Hypothesis testing is a statistical method used to determine whether there is sufficient evidence in sample data to support or reject a claim about a population.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

A. The null hypothesis (H_0) is the statement that there is no significant effect, difference, or relationship. The alternative hypothesis (H_1) is the statement that there is a significant effect, difference, or relationship.

For example:

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

50. How can a confidence interval be used to make a decision about a hypothesis?

A. A confidence interval can be compared with the value stated in the null hypothesis.

For example, suppose:

$$H_0: \mu = 50$$

and a 95% confidence interval for the population mean is (52, 58).

Since 50 is not inside the confidence interval, we reject H_0 at the 5% significance level.

If the hypothesized value is inside the confidence interval, we generally fail to reject H_0 at that significance level.

Conclusion: Confidence intervals provide both an estimate and information about the uncertainty of that estimate, making them useful for statistical decision-making.